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Some reflections on the role of evidence in improving education

Introduction

As Adrian Simpson notes in his introduction to this special issue, the idea that educational policy should be based on clear evidence of “what works”, rather than prejudice, or whim, is obviously attractive, and the basic underlying logic is sound. After all, there can be few people advocating changes that will have no impact on education, or even worse, will actually lower student achievement.

Once we accept that evidence from carefully conducted research studies can help in the improvement of educational outcomes, the question that immediately follows is “what kinds of evidence”, and in recent years, the idea that particular kinds of research studies – and in particular, systematic reviews and randomised control trials – should be the main, or only source of evidence for such efforts has received a great deal of support, at least from some quarters.

However, as the contributors to this collection point out, the particular claims made by many proponents of such “evidence-based education” – that evidence should be derived largely or only from randomised control trials and meta-analyses – do not hold water. Even if the approaches they advocate are adopted, they will not achieve the ends they seek. Even if we accept that randomised control trials are the best way to produce reliable knowledge about how to raise student achievement, right now, the number of such trials is so small that they cannot provide useful insights on a range of important issues. As Robert Slavin asked many years ago, “Do we really know nothing until we know everything?” (Slavin, 1987, p. 347). More importantly, as several papers in this collection point out, even if we had more, and better, randomised control trials, these would, by themselves, provide little basis for making educational policy or assisting in instructional decision making.

At the other end of the spectrum are those that doubt that educational research has anything much to offer teachers beyond either bland platitudes or things that might work in controlled settings but cannot work in “real” classrooms. While such extreme positions are not common, there are many researchers, practitioners, and policymakers who accept that educational research can be useful, but that it is inappropriate to regard randomised control trials and systematic reviews as inherently superior. Proponents of evidence-based education, they argue, want to treat teachers as technicians rather than as professionals. Teachers are reduced, in Lawrence Stenhouse’s (1980) memorable phrase, to “intellectual navvies” who are told “where to dig their trenches without having to know why” (p. 5). The idea that teachers should be professionals rather than technicians is attractive, but for me this is an empirical question: Is student achievement (however we choose to define it) improved if we let teachers follow their professional judgement without paying any attention to research evidence? Or can research evidence, whether from randomised control trials, such as systematic reviews of research (including meta-analyses), and other kinds of inquiries help inform our efforts to improve student outcomes? And if they can, do randomised control trials and systematic reviews (including meta-analysis) have a privileged role, or do they form one source of evidence among others?

The challenge, then, is to clarify the role that evidence from randomised control trials and systematic reviews might play in the search for ways of improving educational achievement, and the papers in this special issue provide a number of helpful, complementary, perspectives on this issue. I am aware that in a closing reflection such as this, it is conventional to offer some critique of the papers, highlighting potential weaknesses in the arguments, and offering contrary perspectives, but to be honest, I found so little to disagree with, that this would be a very short piece.

Instead, I offer a number of reflections on the arguments made in the papers, drawing out some themes, and offering some suggestions about how evidence might make more productive contributions to the improvement of education. I begin by reviewing briefly some of the background to the current interest in evidence-based education, and then focus in more depth on the particular claims made for randomised control trials in education. I conclude with some suggestions about how the ideas presented in this special issue might contribute to helping all stakeholders in the educational process become more critical consumers of educational research.

Background

In her “troubling” history of educational research, Ellen Condliffe Lagemann (2000) suggests that one way to understand this history is as a search for disciplinary foundations. Towards the end of the 19th century, it was thought that the emerging science of psychology might provide clear guidance about the best way to educate young people. In the early decades of the 20th century, the founders of the School Survey Movement argued that answers could be found in the “big data” of the time, by collecting detailed information on curriculum, school finance, graduation rates, instructional time, and so on. The successful administration of the Army Alpha test, to almost 2 million army recruits in the closing year of World War I (Zenderland, 1998), led some to argue for the importance of testing in efforts to raise educational achievement, while others thought that studies of curriculum and instruction were more likely to yield insights into the best ways of improving education.

In the second half of the 20th century, additional perspectives, from sociology, anthropology, computer science, and a number of other disciplines, were added to the mix, and while these have undoubtedly enriched the ways that educational processes are viewed, the role of research in the improvement of education remains a vexed, and vexing, issue.

For some, of course, these multiple perspectives are just the result of researchers making things more complex than they need to be – perhaps best summed up by Senator Robert F. Kennedy’s questioning of Commissioner of Education Harold “Doc” Howe in 1966: “Do you mean that you spent a billion dollars and you don’t know whether they can read or not?” (Wickline, 1971, p. 7). What was needed instead, many politicians and policymakers argued, was a need for a simple focus on “what works” in education. This culminated, in 2002, in the creation of the “What Works Clearinghouse” (WWC) that was intended to be “a central and trusted source of scientific evidence for what works in education” (WWC, 2011, p. 1).

Across the Atlantic Ocean, in the mid-1990s, the national school inspection agency in England (the Office for Standards in Education, or Ofsted) had suggested that it would be useful for teachers to have access to authoritative reviews of research in particular fields, and had commissioned a series of reviews of research for teachers (Arnot, Gray, James, Rudduck, & Duveen, 1998; Askew & Wiliam, 1995; Gillborn & Gipps, 1996; Reynolds & Farrell, 1996). The idea that educational research could assist in educational improvement was given further impetus by David Hargreaves in the Teacher Training Agency’s Annual Lecture in 1996 (Hargreaves, 1996).

In his speech, Hargreaves (1996) compared education unfavourably with medicine in the way that research informed professional practice, and while he explicitly rejected the idea that education should copy medicine, he argued that education should seek, like medicine, a more productive relationship between research and professional practice.

Over the next few years, the idea that educational policy should be informed by systematic reviews of research evidence gained momentum. In 1998, Robert Marzano published a “theory-based meta-analysis of research on instruction” that sought to synthesise all the relevant research on the factors that impacted the learning, and the following year, the Campbell Collaboration was founded, aiming to bring to the social sciences the approaches that the Cochrane Collaboration had developed for medicine. More recently, John Hattie’s “Visible Learning” project (Hattie, 2009) has sought to synthesise all the influences on student achievement in order to provide clear guidance for teachers and policymakers about the kinds of innovations that are most likely to improve educational outcomes.

Cowen (this special issue) describes how the use of such evidence is useful in solving the “bureaucratic knowledge problem” – the difficulty that bureaucrats have in knowing which evidence to trust in their policymaking, and in knowing whether practitioners are faithfully implementing the policies that have been adopted. This issue was particularly salient in the United Kingdom in 2010, because although the Conservative Party won more parliamentary seats than any other party in the general election held in May of that year, to gain an overall majority they needed to enter into a coalition with Liberal Democrat members of parliament. Participating in this coalition required the Liberal Democrats to abandon some long-held policies, such as the abolition of tuition fees in higher education, and so they were anxious to show that other policies, such as additional funding for at-risk students (the “pupil premium”), were being implemented.

While schools serving at-risk students did receive additional funding via the pupil premium, the way that schools received these funds made it impossible to determine how they were being used, and in particular, whether they were being used as intended, namely, for improving the achievement of at-risk students. The creation of the “Pupil Premium Toolkit” in 2011 – a synthesis of research on the effects of educational interventions, especially those intended or believed to be more effective for at risk students – provided what Cowen describes as “an additional strategy for sovereign delegation”. As Cowen notes, the use of systematic reviews of randomised control trials, and in particular an “approved menu” of interventions as given in the Toolkit, provides a convenient mechanism for both aspects of the “bureaucratic knowledge problem”. Policymakers can ensure that the policies being recommended are ones of which they approve, and the reduction of all possible interventions to a relatively short list makes it much easier to monitor whether schools are, in fact, implementing the favoured interventions. The question then, of course, is what kinds of research should be so favoured.

What kind of evidence improves education?

As each of the papers in this special issue point out, there are a range of perspectives on the role that educational research might play in improving education, and one particular issue – the use of randomised control trials – has gained particular prominence in debates amongst both policymakers and researchers.

At one extreme are those who argue that only evidence generated by randomised control trials can be taken seriously, and that any evidence arising from other kinds of studies is problematic or even irrelevant. At the other extreme are those who regard the entire idea of evidence-based education as a kind of category error – ascribing to something a property it cannot have, like describing a rock as happy.

Neither of these extreme positions, it seems to me, is tenable, and so we have to look at more moderate and nuanced positions between these two extremes – adopting what Phillips (this special issue) describes, adopting William James’ terms, the “tough-minded” and the “tender-minded” camps.

As several of the papers in this special issue illustrate clearly, the claims made for evidence-based education, and in particular the role that randomised control trials might play within such an approach, do not stand up to scrutiny even in their own terms. Wrigley and McCusker (this special issue) point out that what is often described as “scientific research in education” (see, e.g., Shavelson & Towne, 2002) is not particularly “scientific” in that it bears little resemblance to how knowledge is created in the physical and life sciences. Moreover, Joyce (this special issue) points out that implementing randomised control trials is challenging in any setting because participants change their behaviour. For example, in an interview with Russ Roberts (2016), economist James Heckman mentions a trial of a drug that was hoped might be effective in preventing or delaying the onset of AIDS. The trial was blinded – participants did not know whether they were getting the drug or a placebo – so they shared their assigned medications with others in the trial to maximise the chance of getting some of the drug, rendering the results difficult, if not impossible, to interpret. In social science settings, such as child care and education, blinding is much more difficult, and often impossible. Heckman points out that in randomised control evaluations of the effects of *Head Start*, parents whose children were not assigned to receive *Head Start* placements knew it, and made other arrangements, in some cases securing higher quality child care than was being provided in *Head Start*. They had been attracted to the trial by the prospect of getting free or reduced-cost child care, knowing full well that if they were assigned to the control condition, they were free to secure high-quality child care in other ways. Furthermore, as Joyce (this special issue) points out, even when randomised control trials are well designed and implemented faithfully, generalisations to settings that were not included in the trial are warranted only if the settings in which the research was carried out are representative of others, which is impossible to establish. In other words, we can never know “what works”, but only “what worked” and, as Cartwright (this special issue) points out, we have known for centuries from the work of David Hume (1748/1975) that “there is no rational argument to infer knowledge of the unobserved from knowledge of the observed”.

In response, advocates of evidence-based education often argue that while the kinds of issues raised by the contributors to this special issue may be of theoretical interest, they have little or no practical significance. The factors that critics worry about will not arise in real settings, or, if they do, they will either cancel each other out, or have little impact on the final results. It may therefore be useful to examine these issues in the context of an issue of a real, substantive issue – the impact of class-size reduction programmes on student achievement.

Research on class-size reduction programmes

One of the best designed class-size reduction programmes, and almost certainly the most thoroughly investigated and cited, is the “Student-Teacher Achievement Ratio” (STAR) project carried out from 1985 to 1989 in the US state of Tennessee. This study met many of the requirements typically required of a randomised control trial (with the obvious exception of blinding) and was described by Frederick Mosteller – a leading statistician of the last century – as “one of the most important educational investigations ever carried out” (Mosteller, 1995, p. 113). Kindergarten students were randomly assigned to be in small classes (13 to 17 students), large classes (22 to 26 students), or large classes with a teacher aide, and teachers

were randomly assigned to groups (an important consideration in the United States, where in many districts, teachers with seniority are allowed to select, or at least express preferences for, teaching assignments).

By the end of second grade, students in the smaller classes were 3 months ahead of those in the larger classes, with gains of up to twice as much for students from poorer homes, and minorities (Grissmer, 1999; Molnar et al., 1999), and the benefits seemed to have lasted when students were placed in regular-sized classes from fourth grade on. Levin, Belfield, Muennig, and Rouse (2007) estimate that being placed in smaller classes from Kindergarten through third grade would increase high school graduation rates by 11 percentage points.

The success of the Tennessee STAR study in raising student achievement (admittedly at very high cost) led a number of other states, such as California, to introduce similar class-size reduction programmes, but these were generally much less successful (see, e.g., Stecher & Bohrnstedt, 2002). The fact that the promising results of the Tennessee STAR project failed to generalise to other contexts provides a stark illustration of the issues raised by the contributors to this special issue.

First, as Cartwright and Joyce point out in their contributions to this special issue, to generalise, it is necessary to assume the schools that participate in a study are representative of the schools that do not, and this was clearly not the case in the Tennessee STAR project. In a number of respects, the schools participating in the study were systematically different from the average for the state. For example, the schools participating in the study were, on average, larger, because to be eligible for the study, each school needed approximately 60 students in each grade (in order to create the two large classes and one small class that the study required). Hanushek (1999) reports that the proportion of African American students in the participating schools was over 40% higher than the average for the state. Given that smaller classes appeared to be more beneficial for students from minority groups, smaller effects would be expected were class-size reduction programmes implemented in schools with a smaller proportion of minority students.

Second, as Joyce (this special issue) reminds us, making causal claims from the results of randomised control trials requires that randomisation is maintained. It appears that when some parents found out their child had been allocated to a larger class, they brought pressure on the school to move their child to one of the smaller classes. Whatever the reason, the randomisation protocols that were used in the Tennessee STAR study did not result in random allocation of students to classes. Particularly in the schools that were more effective on average, more affluent students were over-represented in the smaller classes (Sohn, 2015). In other words, the increased achievement observed in the smaller classes may have been at least in part caused by a migration of higher achieving students to the smaller classes. Furthermore, there were substantial changes in the composition of the classes over time, and it appears that missing data were more common in the large than in the smaller classes.

Third, at the time the STAR results were published, there was little firm evidence about the variability of teacher quality, although it is worth noting that as early as 1971, Eric Hanushek had shown that the available data were not consistent with the assumption that all teachers were equally effective (Hanushek, 1971). The Tennessee STAR study was relatively small in scale, requiring an additional 50 teachers, and it seems reasonable to assume that the additional teachers required as a result of the study were as effective as those already working in the schools. However, when class-size reduction programmes are implemented state-wide, this is unlikely to be the case. In California, for example, when a state-wide class-size reduction programme was introduced in 1996, the number of uncertified teachers working in California schools increased rapidly, from around 1 in 200 in 1990 to around 1 in 8 in 1997 (Jepsen & Rivkin, 2002). Moreover, new and uncertified teachers were not allocated

randomly across the state. As Randy Ross (1999) pointed out, many experienced teachers left urban schools to take up newly-created positions in more affluent areas. The net effect of class-size reduction in California was to reduce average teacher quality across the state, and to concentrate those reductions in the schools serving the most disadvantaged students.

This case study of class-size reduction programmes also provides an illustration of another of the themes raised by the articles in this special issue and that is related to what Kvernbekk (this special issue) calls “Tales not told”. Limiting our educational innovations to those that are supported by randomised control trials, we are likely to miss out on many things that might help. As Phillips (this special issue) points out, there are many issues, such as whether parachutes are effective, where randomised control trials cannot be carried out (see also Yeh et al., 2018).

More seriously, if randomised control trials are the only way of producing knowledge, then we do not know that smoking causes lung cancer. But the “Tales not told” include not only those cases where there are no randomised control trials, but also those where there is little or no evidence because the kinds of research that would have provided useful evidence have not been carried out.

Reviews of research on class-size reduction programmes typically conclude that such programmes are more effective for younger students, and less effective – and sometimes that they are completely ineffective – for older students. However, such reviews rarely consider why this might be. One obvious explanation relates to the ways that teachers teach students of different ages. In elementary school, particularly in the early grades, teachers spend a great deal of time working with individual children or with small groups, and so it is easy to see how smaller classes would result in more individual attention for each student, with a consequent increase in achievement. In secondary schools, it is common to find teachers working with a whole class, and a teacher talking at a class of 20 is unlikely to be significantly more effective than a teacher talking at a class of 30. However, if class-size reduction programmes were combined with inservice support for teachers to help them develop teaching approaches that were only possible with smaller classes, the results might be different. In such a case, the changed methods of teaching can be thought of as the cause of the improved achievement, with the reduction of class size functioning as what Cartwright and Hardie (2012) describe as “support factors”. The problem is that these kinds of research studies are all but non-existent, because they are even more expensive than regular class-size reduction programmes. The important point here is that educational research tells us only what was, not what might be.

Another example where the failure to consider issues of teacher quality means that findings are difficult, if not impossible, to interpret is provided by research on ability grouping.

Research on ability grouping

Researchers and policymakers have debated for years about whether students learn better when they are taught in groups with a restricted range of achievement or ability. The Education Endowment Foundation (EEF), in its Teaching and Learning Toolkit, concludes that the available evidence suggests that such “ability grouping” is not effective, and may in fact slightly lower student achievement (EEF, 2018). Of course, almost all of these studies are cross-sectional in nature, so we might decide that what is needed is a randomised control trial. We would recruit schools to the study, and then randomly assign the schools to teach their students either in heterogeneous or in homogeneous groups. Even if schools adhered to the grouping practices they were asked to follow, what we might learn from such a study is limited because of the effects of teacher quality.

As noted above, it has become clear in recent years that some teachers are much more effective than others. While estimates of the magnitude of teacher effects vary, a large body of evidence suggests that, at least in the United States, when measured with standardised achievement tests, one standard deviation of teacher quality is associated with an increase in student achievement of around 0.15 standard deviations (Hanushek & Rivkin, 2010). One way to illustrate what this means is that in a group of 50 teachers, students taught by the most effective teacher in that group will learn in 6 months what students taught by the average teacher will take a year to learn, and those taught by the least effective teacher in that group of 50 will take 2 years to make the same progress. In addition, more effective teachers are disproportionately more effective for lower achieving students (Slater, Davies, & Burgess, 2012). What this means is that if we conduct a randomised control trial, where some schools are selected, at random, to group students by ability, and others to teach students in mixed-ability groups, and we find that average student achievement is lower when students are grouped by ability, we cannot conclude that it is the grouping by ability that has caused this lowered achievement. It could simply be that in the schools that were assigned to teach using ability grouping, the most effective teachers were assigned to teach the higher achieving students. Students who would have benefitted most from being taught by the most effective teachers would be less likely to be taught by effective teachers than similar students in the schools using mixed-ability teaching. In other words, what we would be observing could be a teacher-quality effect. In this respect, it is worth noting that the EEF's current study into the effects of ability grouping for the teaching of mathematics to 11- and 12-year-olds (EEF, 2019) has not used random assignment, because it was impossible to get schools to agree to this, and has no direct measures of teacher quality, so it will not be possible to determine whether any effects on student achievement were caused by the form of grouping used, or teacher quality. Given that, as noted above, the magnitude of between-teacher effects is likely to be far larger than any differences caused by grouping practices, it is hard to see what light such a study would shed on the issue at hand.

On a personal level, this is particularly sobering, because I spent 5 years on a study of grouping practices in mathematics in six schools in the South East of England. In the project, my colleagues and I tracked around 1,000 students over 5 years, as they moved from being taught in mixed-ability groups when they began secondary school to being taught in more or less homogeneous groups as they got older. Because some schools introduced ability grouping earlier than others, we were able to investigate the impact on student achievement, using what econometricians call a difference-in-differences approach (Angrist & Pischke, 2015) and showed that grouping students by ability lowered achievement (Boaler, Wiliam, & Brown, 2000; Wiliam & Bartholomew, 2004). Or so we thought.

At the time we conducted those studies, we were not aware of the magnitude of between-teacher effects, and so did not collect any evidence on teacher quality. Now that evidence shows that between-teacher effects are substantial, it is clear that the conclusions of our study are not so clear cut. It could be that it was the way that teachers were allocated to classes that caused the differences in student achievement, and not the ability-grouping practices themselves.

What is more important, in terms of the arguments for and against evidence-based education being discussed in this special issue, is that even if we had been able to conduct a randomised control trial but had not understood the variability of teacher quality, then our results might well be misleading, or just plain wrong. If we did find that grouping by ability was associated with lower average achievement, we would need to look at how teachers were allocated to classes to make sense of our results. And given the research of Slater et al. (2012) discussed above, it would suggest that grouping students by ability, but ensuring that the most effective

teachers were assigned to teach the lowest achieving students, might yield very different outcomes. Again, educational research only tells us what was, not what might be.

The fact that well-designed studies might support particular conclusions at one point in time, but not later on, suggests that criterion of validity, discussed at length in the contributions of Cartwright, Joyce, and Phillips to this special issue, cannot be a property of experiments, or even of their results, but rather of the inferences that we draw from those results.

This is now broadly accepted in the field of psychometrics. A mathematics test with a high reading demand might be a valid assessment for good readers, in the sense that the test adequately sampled the relevant domain, was strongly related to the construct of interest, and produced similar results to other assessments of the same domain. However, for poor readers, it would be impossible to determine whether low scores were caused by lack of mathematical ability or of an inability to understand the questions. Within the psychometric community, there is broad (although not quite complete) agreement that validity is a property of inferences, not of assessments. As Lee Cronbach pointed out years ago, “One validates, not a test, but an *interpretation of data arising from a specified procedure*” (Cronbach, 1971, p. 447, emphasis in original).

Perhaps the most important consequence of this definitional shift is that, within educational assessment, the emphasis has shifted from the idea of validity as a property of tests, or even inferences, to the idea of validation as a process. In his seminal work on validity, Samuel Messick (1989), argued that validation is a never-ending process of marshalling evidence that, on the one hand, supports the preferred interpretation of an assessment outcome and, on the other hand, also that alternative interpretations are less warranted (a point also emphasised by Joyce, this special issue). The question then, of course, is, if validation is an endless process, how can any claims be established. One possible, and in my view, fruitful, answer comes from work of Alvin Goldman and his work on discrimination and perceptual knowledge (1976).

Discrimination and perceptual knowledge

In a paper titled “Discrimination and Perceptual Knowledge”, Goldman (1976) shows that the traditional view of knowledge as justified true belief is inadequate as a basis for deciding whether someone knows something or not. He gives the example of a man named Henry driving in the countryside and pointing out various objects to his son, such as a tractor, a cow, and a silo. At one point, as they pass a barn, he says, “That’s a barn”, which indeed it is, but what Henry does not know is that in this part of the world, there are a number of facsimiles of barns made of papier mâché, and they are just façades that could not be used as barns. In this particular case, what Henry sees is a barn, he believes it is a barn, and his justification of that true belief is the same whether there are papier-mâché barns in the area or not. And yet, we would be inclined to say that, if there are papier-mâché barns in the area, then he does not know that what he is looking at is a barn, but if there are no such false barns around, then he does know that he is looking at a barn.

Goldman (1976) suggests that a way out of this difficulty is to decide that knowledge requires that an individual is able to distinguish or discriminate between a state of affairs in which something is true from a *relevant* possible state of affairs in which it is false. We would say that Henry does not know that what he is looking at is a barn because we know (even though Henry does not) that there is a relevant alternative state of affairs – that what Henry sees is a facsimile – and Henry is unable to distinguish between these two circumstances.

Goldman’s idea that knowledge involves being able to distinguish between *relevant* alternative states of affairs makes it clear that our knowledge is provisional. To continue with the

example of ability grouping, as new evidence about the importance of teacher quality becomes available, then our inability to distinguish between a state of affairs in which the lower achievement was caused by the effects of teacher quality and a state of affairs in which it was caused by the effects of ability grouping means that we now do not know that ability grouping lowers achievement. Note however, that before the evidence about importance of teacher quality was apparent, we could be said to know this.

Goldman's focus on the relevance of alternative states of affairs also allows that knowledge can be different from locality to locality. Policymakers in areas where teacher recruitment is challenging have knowledge about the likely impact of class-size reduction programmes that would not be relevant to those in areas where the supply of good teachers is plentiful.

Goldman's formulation also provides a powerful way of looking at the various debates around what counts as evidence in education. For advocates of randomised control trials one particular alternative state of affairs – that the two groups being compared were not equivalent at the beginning of the study – is accorded more importance. For the contributors to this special issue, other states of affairs, such as participants in the study not being representative of the wider population, are also important.

To take a specific example, Cowen (this special issue) points out that the EEF's emphasis on feedback as the single most cost-effective intervention justified "additional pressures on teachers from inspectors that are ultimately not productive" even though few, if any, of the studies that the EEF included in its review looked at the effects of marking in school. Furthermore, Tone Kvernbekk (this special issue) points out, in her discussion of four models of how evidence might move people to more effective action in education as several contributors to this special issue point out, "an overall positive average result is compatible with negative results for some of the individuals". In response, advocates of evidence-based education often argue that as long as the average effect is positive, we should not worry too much that there may be some who are harmed, much in the same way that while some people may have been killed in automobile accidents because they were wearing a seatbelt, the average effect of mandating seat belt use has been to save thousands of lives. The problem with such an argument is that without some knowledge of how the intervention has its effects, then applying the intervention in a particular context may produce a *consistently* negative effect.

One particularly careful review of feedback studies by Kluger and DeNisi (1996) covering the period between 1905 and 1995 found that, on average, the effect of feedback in the 131 well-designed studies they identified was to increase achievement by approximately 0.4 standard deviations. The positive average effect of feedback found by Kluger and DeNisi and in his own reviews of research is presumably what led John Hattie to suggest that "the simplest prescription for improving education must be dollops of feedback" (Hattie, 1999). However, the standard deviation of the effect sizes in the studies Kluger and DeNisi included in their review was approximately 1, and around 38% of the 607 effect sizes were negative. In other words, in over one third of the cases they examined, it would have been better simply not to give the feedback. For those who advocate more feedback, the possibility that more feedback might be counter-productive is not a relevant alternative state of affairs – while the effects may be negative for some, the overall benefit is positive. For others, the alternative state of affairs, that the kinds of feedback that lowered achievement may be the kinds of feedback most prevalent in schools, *is* relevant.

What this means, I think, is that the entire project of evidence-based education can never be successful. Any claims about "what works" are necessarily local, in that they are limited to the participants and contexts actually studied and judgement will be needed to apply them in other settings. Moreover, such claims will be provisional, in that new findings, insights, and

understandings will bring previously credible interpretations of research studies into question. If educational research is to contribute to the improvement of practice, teachers, school leaders, and policymakers will have to become critical consumers of educational research.

Helping stakeholders become more critical consumers of educational research

As several of the contributors to this special issue have pointed out, in recent years, there have been a number of attempts to synthesise the available research studies on the influences on student achievement. In one sense, of course, this is laudable, because the idea that one would look to research evidence for guidance about improving education rather than the beliefs of politicians represents a kind of progress. However, as Adrian Simpson points out in this special issue, and elsewhere (Simpson, 2017, 2018), the standardised effect size as defined by Jacob Cohen (1988) is rarely a good indication of the effects even in a single study. Combining those effect sizes in a meta-analysis is fraught with difficulties (William, 2016; Wrigley & McCusker, this special issue), and synthesising the results of a number of meta-analyses by some weighted combination of the effect sizes generated by the individual meta-analyses – what has come to be called meta-meta-analysis – has absolutely no role in educational policymaking. However, it is important to note that Simpson does not “throw out the baby with the bathwater”. He is careful to note that, under certain circumstances, it might be appropriate to combine effect sizes in some way, provided certain conditions are met. In this regard, the list of issues that Wrigley and McCusker (this special issue) provide in their critique of the work of John Hattie is useful. Not only does it serve as a list of what is wrong with meta-meta-analysis. It also serves as a checklist that can be used to see whether the results of a meta-analysis are worth taking seriously:

- Are the studies included relevant?
- Are the effect sizes the results of interventions, or merely associations?
- Are the effect sizes comparing the same things (e.g., comparison with an alternative intervention or “business as usual”, rather than just before and after)?
- Are the effect sizes at the same level (e.g., individual, group)
- Is there a check on publication bias, for example, through the use of a funnel plot?
- Are the interventions being compared of similar duration?
- Are the interventions being compared of similar intensity?
- Are all the outcome measures used all measuring the same thing?
- Is the dispersion of the measured attribute similar in the studies being compared?
- Are the outcome measures used equally sensitive to the treatment being investigated?
- Do the participants in the studies have equal aptitude for the treatment being investigated?

If the answers to all of these questions is “yes”, or at least if the meta-analysis at least examines these issues, for example, by including these issues as moderators of effect, then the meta-analysis may be worth taking seriously. If, however, the answers to any of these questions is no, then it is unlikely that the meta-analysis has much relevance to real educational settings.

Ultimately, however, whatever the advocates of evidence-based education may wish, there can be no objective procedure for identifying and disseminating “what works” in education. Identifying how teachers, schools, districts, and jurisdictions can most profitably direct their efforts will always involve the application of judgement, taking into account local knowledge, the likely relevance of findings from other contexts, the feasibility of proposed reforms (politically, economically, culturally, etc.), and a whole host of other factors.

To provide some structure to aid stakeholders in this messy process, I offer four questions that may help in making smarter decisions about educational priorities:

- (1) *Does this solve a problem we have?* For example, advocates of merit pay for teachers often cite studies from low-income countries where paying teachers for their performance increased student achievement (see, e.g., Duflo, Hanna, & Ryan, 2012). In these countries, getting teachers to turn up for work was a serious problem, with absenteeism rates in excess of 30%. Paying teachers to turn up for work may well help in such situations, but where teachers are already turning up for work (as is the case in most rich countries), such incentives are unlikely to help.
- (2) *How much improvement will we get?* Without some indication of the amount of improvement that an initiative will yield, no sensible decision can be taken about whether it is worth bothering with. In this regard, it is worth noting that reports often cite findings where the fidelity of implementation is high, and while such findings are of interest to researchers, they are less relevant to policymakers. The researcher's question – "Does this work when it is implemented properly?" – is generally less relevant than the policy-maker's question – "If we adopt this intervention, will things get better, and if so, by how much?"
- (3) *How much will it cost?* In his book *Visible Learning*, John Hattie (2009) suggested that effect sizes of less than 0.4 standard deviations were not worth bothering with because that was the average effect of business as usual (although he appears to assume that most quoted effect sizes are gross, rather than, as is more common, net of "business as usual"). However, small effect sizes can be very helpful if they are relatively easy to secure at modest cost. A good example of this is the READY4K! programme, which sends text messages to parents of pre-school children three times a week to remind them to undertake literacy activities with their children. While the effects were small, the modest cost – approximately \$10 per child – makes it highly cost effective.
- (4) *Will it work here?* As discussed earlier, class-size reduction programmes can be effective, especially for at-risk students, provided the additional teachers being recruited are as good as the ones already in post, and provided teachers take advantage of the additional possibilities of smaller classes. In areas where teacher recruitment is challenging, class-size reduction may even reduce student achievement. Only those with what Friedrich Hayek (1945) called "the knowledge of the particular circumstances of time and place" (p. 521) are able to decide which innovations are likely to be effective, and those which, while they might work somewhere else, are likely to be ineffective, or even counterproductive, in that particular setting.

In educational research, "What works" is usually the wrong question because almost anything works somewhere, and nothing works everywhere. A better question is, "Under what circumstances does this work", which is why using research to improve education cannot be achieved by slavishly following a recipe dictated from what Cowen (this special issue) calls a "sovereign". Instead, stakeholders need to become critical consumers of educational research, and the contributions to this special issue provide a wealth of useful guidance about how this might be done.

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